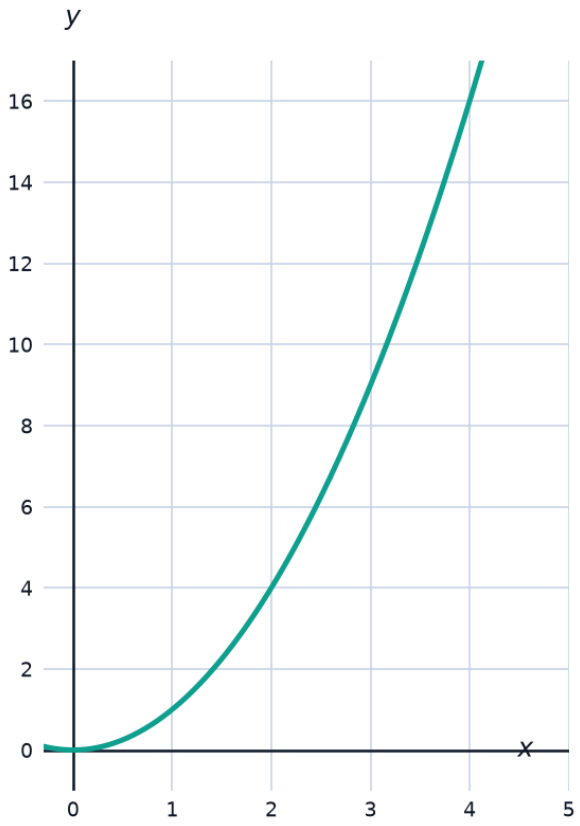
Riemann Sums and Definite Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Approximating area with Riemann sums

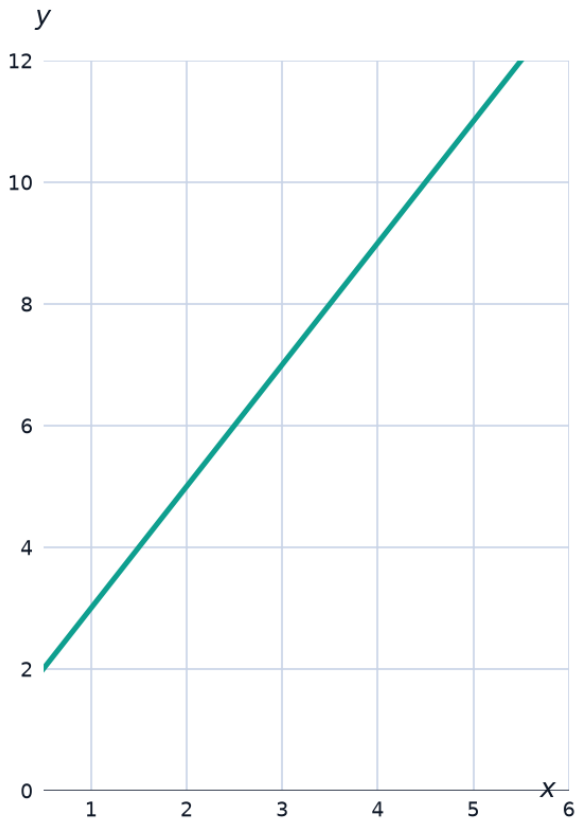
Estimate $\int_0^4 x^2\,dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.



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2 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_1^5 (2x+1)\,dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.



3 [Foundational] • Target: The definite integral and its properties

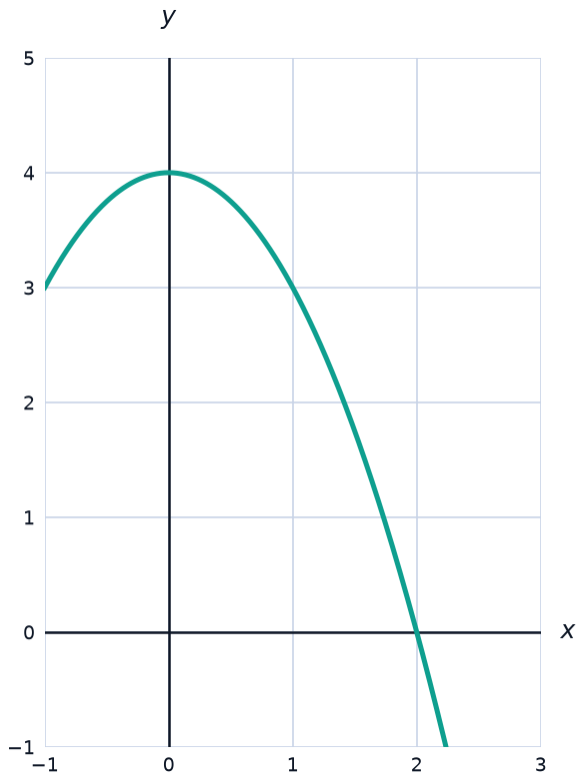
Given $\int_0^3 f(x)\,dx = 8$ and $\int_3^7 f(x)\,dx = -2$, find:

a $\int_0^7 f(x)\,dx$ **b** $\int_7^3 f(x)\,dx$ **c** $\int_0^3 5f(x)\,dx$

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4 [Foundational] • Target: The definite integral and its properties

The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2)\,dx$ exactly.



5 [Foundational] • Target: The definite integral and its properties

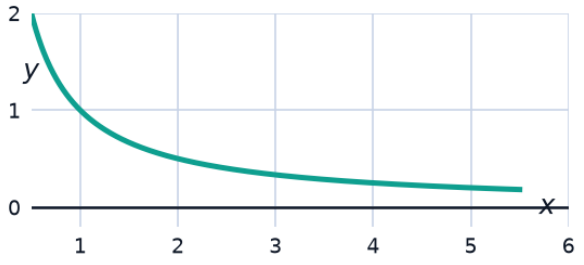
Evaluate $\int_{-2}^{2} (x^3 - 4x)\,dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

6 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_0^4 x^2\,dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.

7 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.

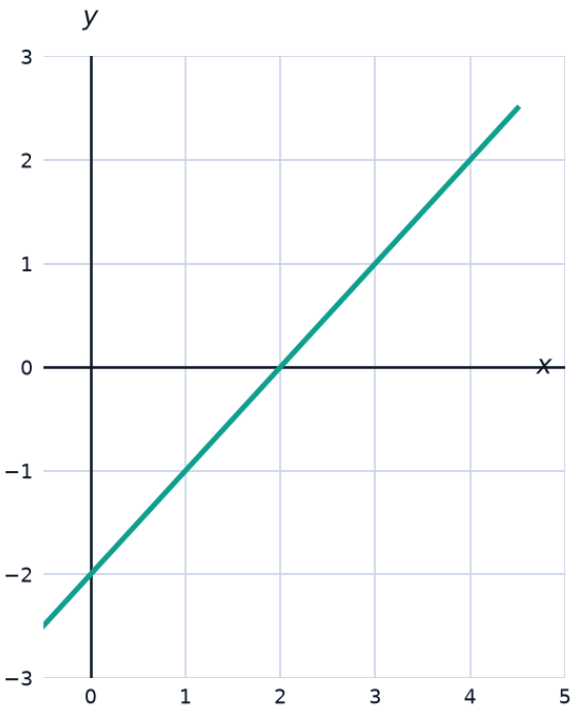


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Signed area**

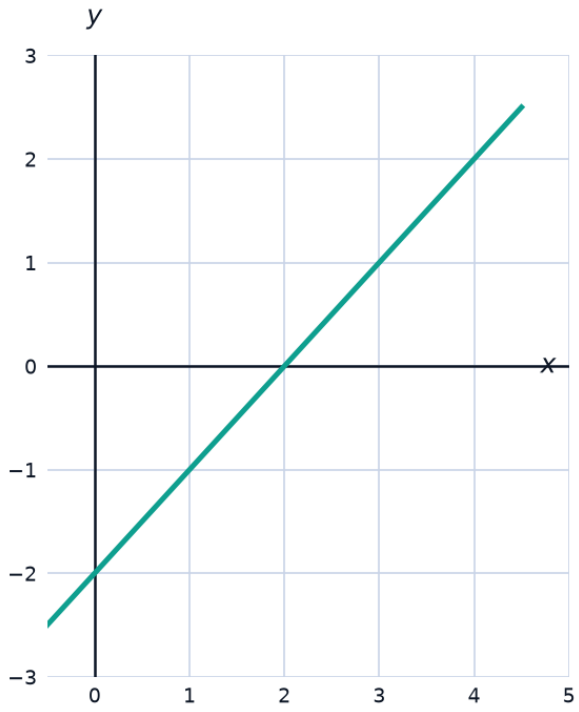
The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x - 2)\,dx$ and explain why the result does not equal the total area between the curve and the axis.



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9 **[Intermediate]** • **Target: Signed area**

Find the total (unsigned) area between $f(x) = x - 2$ and the x-axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.



10 **[Intermediate]** • **Target: Comparing Riemann sum estimates**

For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

11 **[Intermediate]** • **Target: Comparing Riemann sum estimates**

Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

12 **[Intermediate]** • **Target: Riemann sums from a table of values**

A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

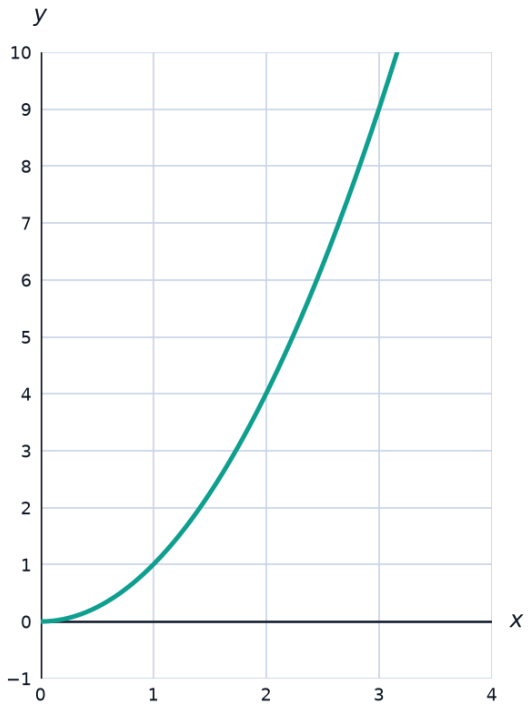
13 **[Intermediate]** • **Target: The definite integral as a limit of Riemann sums**

Write the formal limit definition of the definite integral $\int_a^b f(x)\,dx$ in terms of a Riemann sum, and explain the role of $n \to \infty$.

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14 [Intermediate] • Target: Riemann sums with sigma notation

Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2\,dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

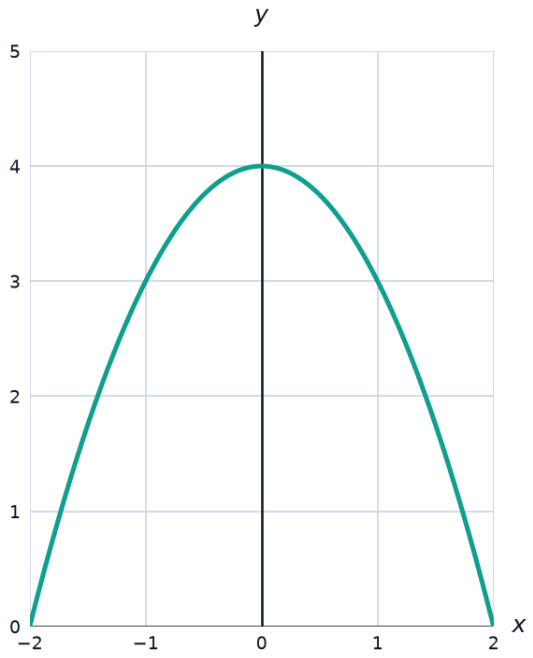
15 [Challenging] • Target: Riemann sums with sigma notation • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Estimate $\int_0^6 f(x)\,dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.

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16 **[Challenging]** • **Target: Comparing approximation methods on a concave function** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.



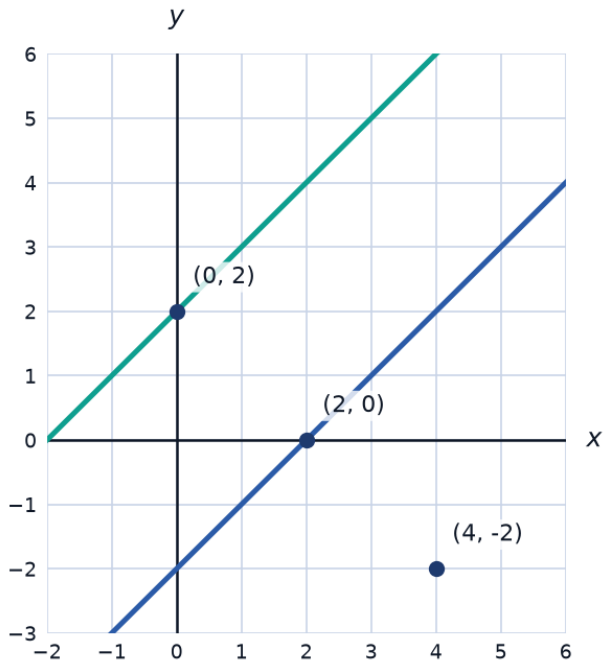
17 **[Challenging]** • **Target: Signed area** • **Cross-Lesson: 15.01 Double Integrals**

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:.

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18 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$? Show your complete algebraic work.



19 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 2.01 Evaluating Limits

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$? Show your complete algebraic work.

20 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 2.02 Continuity

Evaluate the improper integral $\int_1^\infty \frac{\ln(x)}{x^4}\,dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Left and Right Riemann Sums

Approximate $\int_0^4 x^2\,dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14

b. 30

c. 21

d. 16

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22 [Foundational] • Target: Left and Right Riemann Sums

Approximate $\int_0^4 x^2\,dx$ using a Right Riemann sum with $n = 4$ equal subintervals.

a. 30

b. 14

c. 22

d. 28

23 [Foundational] • Target: Left and Right Riemann Sums

For a strictly increasing function $f(x)$ on $[a, b]$, which inequality correctly relates the exact integral I to the Left (L_n) and Right (R_n) Riemann sums?

a. $L_n < I < R_n$

b. $R_n < I < L_n$

c. $I < L_n < R_n$

d. $L_n = I = R_n$

24 [Foundational] • Target: Midpoint and Trapezoidal Approximations

Approximate $\int_1^5 \frac{1}{x}\,dx$ using the Midpoint Rule with $n = 2$ equal subintervals.

a. $\frac{1}{2} + \frac{1}{4} = 0.75 \approx 1.5$ is wrong; width=2: $2(1/2 + 1/4) = 1.5$

b. 1.5

c. 1.2

d. 1.33

25 [Foundational] • Target: Midpoint and Trapezoidal Approximations

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^2 x^3\,dx$.

a. 4.25

b. 4.00

c. 3.75

d. 4.50

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26 [Foundational] • Target: Midpoint and Trapezoidal Approximations

If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal approximation T_n compare to the exact integral I?

a. $T_n > I$ (overestimate)

b. $T_n < I$ (underestimate)

c. $T_n = I$

d. Cannot be determined without $f'(x)$

27 [Foundational] • Target: Definite Integral as the Limit of a Riemann Sum

Express the limit $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^4 \frac{3}{n}$ as a definite integral.

a. $\int_2^5 x^4\,dx$

b. $\int_0^3 (2+x)^4\,dx$

c. $\int_2^5 (2+3x)^4\,dx$

d. $\int_0^1 (2+3x)^4\,dx$

28 [Foundational] • Target: Definite Integral as the Limit of a Riemann Sum

Evaluate $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \frac{i^2}{n^2}$.

a. 1/3

b. 1/2

c. 1/4

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Definite Integral as the Limit of a Riemann Sum

Convert $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n+i}$ into a definite integral on $[0, 1]$ and evaluate it.

a. $\ln 2$

b. 1

c. $\ln 3$

d. 1/2

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30 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate $\int_{-3}^{3} \sqrt{9 - x^2}\, dx$ using geometry.

a. $\frac{9\pi}{2}$

b. 9π

c. $\frac{9\pi}{4}$

d. 18

31 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate $\int_{0}^{4} (2x - 3)\, dx$ by computing net signed area.

a. 4

b. 8

c. 0

d. -2

32 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

If $\int_{1}^{5} f(x)\, dx = 7$ and $\int_{3}^{5} f(x)\, dx = 4$, find $\int_{1}^{3} f(x)\, dx$.

a. 3

b. 11

c. -3

d. $7/4$

33 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate the improper integral $\int_{0}^{\infty} \frac{1}{(x+4)^4}\, dx$.

a. $\frac{1}{192}$

b. $\frac{1}{96}$

c. ∞

d. $\frac{193}{192}$

34 [Intermediate] • Target: Approximating area with Riemann sums

Evaluate the improper integral $\int_{1}^{\infty} \frac{\ln(x)}{x^3}\, dx$.

a. $\frac{1}{2}$

b. ∞

c. $\frac{1}{4}$

d. $\frac{5}{4}$

35 [Intermediate] • Target: The definite integral and its properties

Evaluate the improper integral $\int_{0}^{1} x^3 \ln(x)\, dx$.

a. $-\frac{1}{8}$

b. $-\frac{1}{16}$

c. ∞

d. $\frac{15}{16}$

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36 [Intermediate] • Target: Midpoint and trapezoidal approximations

Evaluate the improper integral $\int_0^1 x^2 \ln(x)\,dx$.

a. $-\frac{2}{9}$

b. $-\frac{1}{9}$

c. ∞

d. $\frac{8}{9}$

37 [Intermediate] • Target: Signed area

For $|x| < 5$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{5}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{5}{5-x}$

c. $\frac{5}{x+5}$

d. $\ln(x+1)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 16.01 Vector Fields and Line Integrals

For $|x| < 3$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{3}{x+3}$

c. $\frac{3}{3-x}$

d. $\ln(x+1)$

39 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 2.01 Evaluating Limits

For $|x| < 2$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{2}{x+2}$

c. $\ln(x+1)$

d. $\frac{2}{2-x}$

40 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 2.02 Continuity

For $|x| < 4$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{4}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{4}{x+4}$

c. $\frac{4}{4-x}$

d. $\ln(x+1)$

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41 [Challenging] • Target: Riemann sums with sigma notation • Cross-Lesson: 3.01 The Derivative from First Principles

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞

c. π

d. 2

42 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. True

c. Only for separable functions

d. Depends on Fubini

43 [Challenging] • Target: Left and Right Riemann Sums • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{2^{-n}(x-1)^n}{n^2}$.

a. $(-1, 3)$

b. $[-1, 3]$

c. $[-1, 3)$

d. $(-1, 3]$

44 [Challenging] • Target: Midpoint and Trapezoidal Approximations • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{4^{-n}(x-3)^n}{n^2}$.

a. $(-1, 7)$

b. $[-1, 7]$

c. $[-1, 7)$

d. $(-1, 7]$

45 [Challenging] • Target: Definite Integral as the Limit of a Riemann Sum • Cross-Lesson: 6.01 Related Rates

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x+3)^n}{n}$.

a. $(-4, -2)$

b. $[-4, -2]$

c. $(-4, -2]$

d. $[-4, -2)$

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